

From zero to a running VM

Every screen of one complete lifecycle, shot on a clean **Lubuntu 22.04** desktop: clone the repository, run the setup, open the web dashboard, then **download, install, use, save and stop an Ubuntu 26.04 desktop VM without typing a single vmctl command**. Each step also gives the equivalent command, for the terminal or a script.

The profile is `ubuntu-26.04`. The screenshots are from `vmctl 0.10`: the dashboard opens on the whole catalog because nothing is chosen yet (see *My VMs* in the README once you have favourites).

Times are the ones measured for these screenshots: a Lubuntu VM with 6 vCPUs and 12 GB of RAM, itself running under KVM, on a fast connection. A physical host is faster.

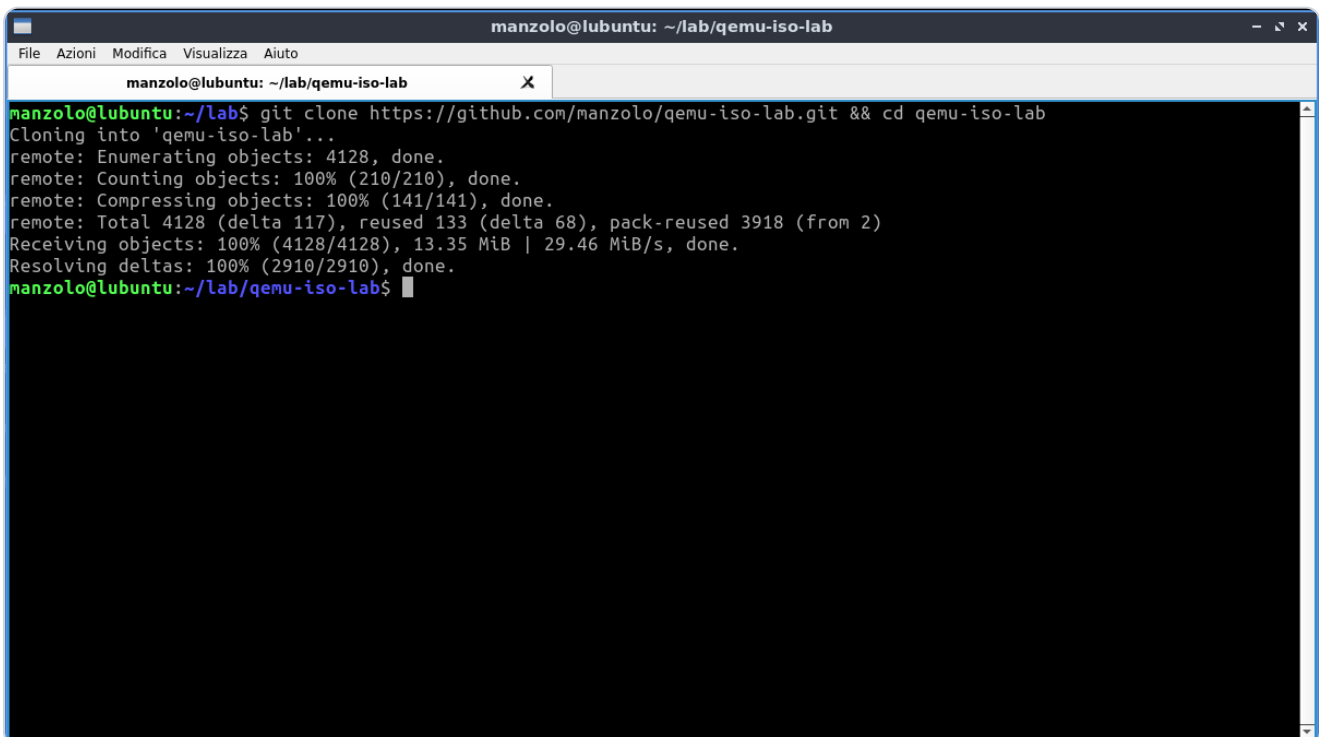
Steps: [1. Get the code](#) · [2. Set up the host](#) · [3. Open the dashboard](#) · [4. Pick a machine](#) · [5. Download the ISO](#) · [6. Install it, unattended](#) · [7. Watch the install](#) · [8. Use the desktop](#) · [9. SSH](#) · [10. Save it, stop it](#) · [What is on disk](#)

1. Get the code

For the shortcut, use the [one-command installer for Linux or Windows](#), then continue at [step 3](#) (on Linux, first `cd ~/qemu-iso-lab`). The screenshots below show the equivalent manual steps.

Open a terminal. Only `git` and `python3` are needed; `make` is not.

```
mkdir -p ~/lab && cd ~/lab
git clone https://github.com/manzolo/qemu-iso-lab.git && cd qemu-iso-lab
```

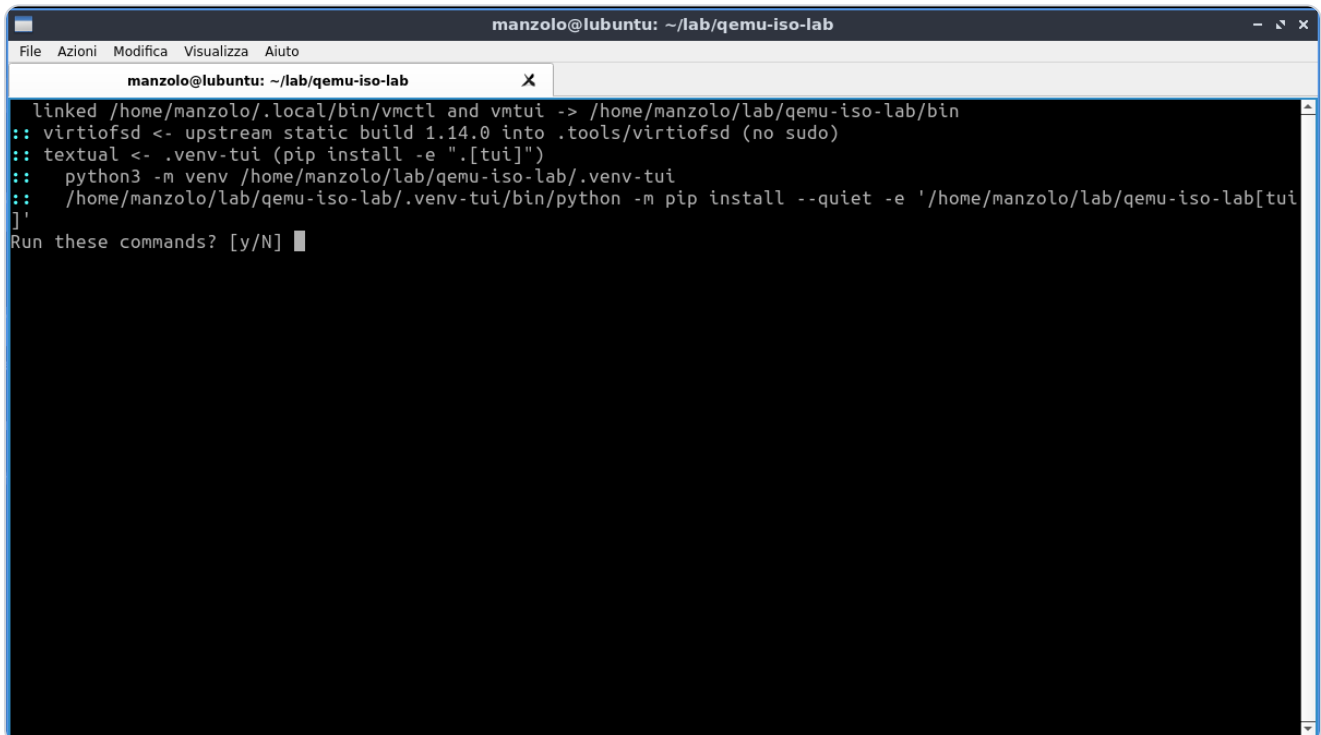
A screenshot of a terminal window titled "manzolo@lubuntu: ~/lab/qemu-iso-lab". The terminal shows the execution of the command `git clone https://github.com/manzolo/qemu-iso-lab.git && cd qemu-iso-lab`. The output displays the progress of cloning the repository, including enumerating, counting, and compressing objects, and receiving the data. The final prompt is `manzolo@lubuntu:~/lab/qemu-iso-lab$`.

```
manzolo@lubuntu:~/lab$ git clone https://github.com/manzolo/qemu-iso-lab.git && cd qemu-iso-lab
Cloning into 'qemu-iso-lab'...
remote: Enumerating objects: 4128, done.
remote: Counting objects: 100% (210/210), done.
remote: Compressing objects: 100% (141/141), done.
remote: Total 4128 (delta 117), reused 133 (delta 68), pack-reused 3918 (from 2)
Receiving objects: 100% (4128/4128), 13.35 MiB | 29.46 MiB/s, done.
Resolving deltas: 100% (2910/2910), done.
manzolo@lubuntu:~/lab/qemu-iso-lab$
```

2. Set up the host

```
./setup.sh
```

It links `vmctl` and `vmtui` into `~/local/bin`, lists every missing package with the exact commands it will run, and **asks before running them**. Answer `y`; `sudo` asks for your password. QEMU, OVMF, `xorriso` and the other helpers come from `apt`; the `virtiofs` daemon, which Ubuntu 22.04 does not package, is the upstream static build in `.tools/`; the Textual dashboard goes into `.venv-tui` without `sudo`. About 20 minutes on a fresh system, mostly `apt`.



```
manzolo@lubuntu: ~/lab/qemu-iso-lab
File Azioni Modifica Visualizza Aiuto
manzolo@lubuntu: ~/lab/qemu-iso-lab X
linked /home/manzolo/.local/bin/vmctl and vmtui -> /home/manzolo/lab/qemu-iso-lab/bin
:: virtiofsd <- upstream static build 1.14.0 into .tools/virtiofsd (no sudo)
:: textual <- .venv-tui (pip install -e ".[tui]")
:: python3 -m venv /home/manzolo/lab/qemu-iso-lab/.venv-tui
:: /home/manzolo/lab/qemu-iso-lab/.venv-tui/bin/python -m pip install --quiet -e '/home/manzolo/lab/qemu-iso-lab[tui]'
Run these commands? [y/N]
```

It ends with the host check and a welcome screen with the next commands. Each one is alone on its line, so a triple-click copies it. On a first run `~/local/bin` is not on the `PATH` yet, so the commands start with `./bin/`. A new login shell fixes that, or the `export` line it shows.

```
manzolo@ubuntu: ~/lab/qemu-iso-lab
File Azioni Modifica Visualizza Aiuto
manzolo@ubuntu: ~/lab/qemu-iso-lab X
export PATH="$HOME/.local/bin:$PATH"

1 Open the lab in your browser (install, boot, console, SSH)
  ./bin/vmctl web --open

2 Or use the terminal dashboard
  ./bin/vmtui

3 Install a first VM with zero clicks, then log in (Debian, ~10 min)
  ./bin/vmctl bootstrap-preseed debian-server
  ./bin/vmctl shell debian-server

• Every profile, then what is on disk and running
  ./bin/vmctl list
  ./bin/vmctl status

• Your user, SSH key and ISOs: copy the example and edit it
  cp vms/profiles/local.json.example vms/profiles/local.json

• All commands (README.md and docs/ for the rest)
  ./bin/vmctl --help

This screen again: ./bin/vmctl welcome

1 Open the lab in your browser
2 Terminal dashboard
3 Install debian-server with zero clicks (~10 min)
4 List every profile
What now? [1-4, Enter for nothing]
```

`./bin/vmctl welcome` prints this screen again at any time.

3. Open the dashboard

```
./bin/vmctl web --open
```

The server listens on **127.0.0.1 only** and prints a URL with a random token; `--open` opens it in the default browser. Keep the terminal open: Ctrl-C stops the page, never the jobs it started.

```
manzolo@ubuntu: ~/lab/qemu-iso-lab
File Azioni Modifica Visualizza Aiuto
manzolo@ubuntu: ~/lab/qemu-iso-lab X
==> vmctl web: the lab in your browser (127.0.0.1 only)
open    http://127.0.0.1:8765/?token=pAYvL-S_GWV5ueUMSP-3LHlD
:: Jobs started here keep running after Ctrl-C; the TUI shows them too.
```

A fresh Ubuntu 22.04 has no graphical browser: Firefox is a snap that is not installed. Then `vmctl web --open` says so. Install one (`sudo snap install firefox`) or open the URL in any browser on the same machine.

4. Pick a machine

Type in the search box (`/` jumps there) and select a profile. The panel on the right is the machine: resources, disk, ISO, SSH address, and the actions that make sense **for its current state**. Right now it has no disk, and the ISO can be downloaded.

The screenshot shows the QEMU ISO Lab interface. At the top, it says "QEMU ISO Lab" with "0 installed · 2 labs" and "Live | vmctl 0.10.0". Below this is a section titled "YOUR LOCAL WORKSPACE" with "Virtual machines" and the subtitle "Build, run and explore your lab." To the right of this section are three summary boxes: "106 Profiles", "0 Running", and "0 With disk". Below the summary is a search bar containing "ubuntu 26" and a filter bar with "My VMs", "All", "With disk", "Running", "Labs", and "Commands F10". The main area displays a table of profiles:

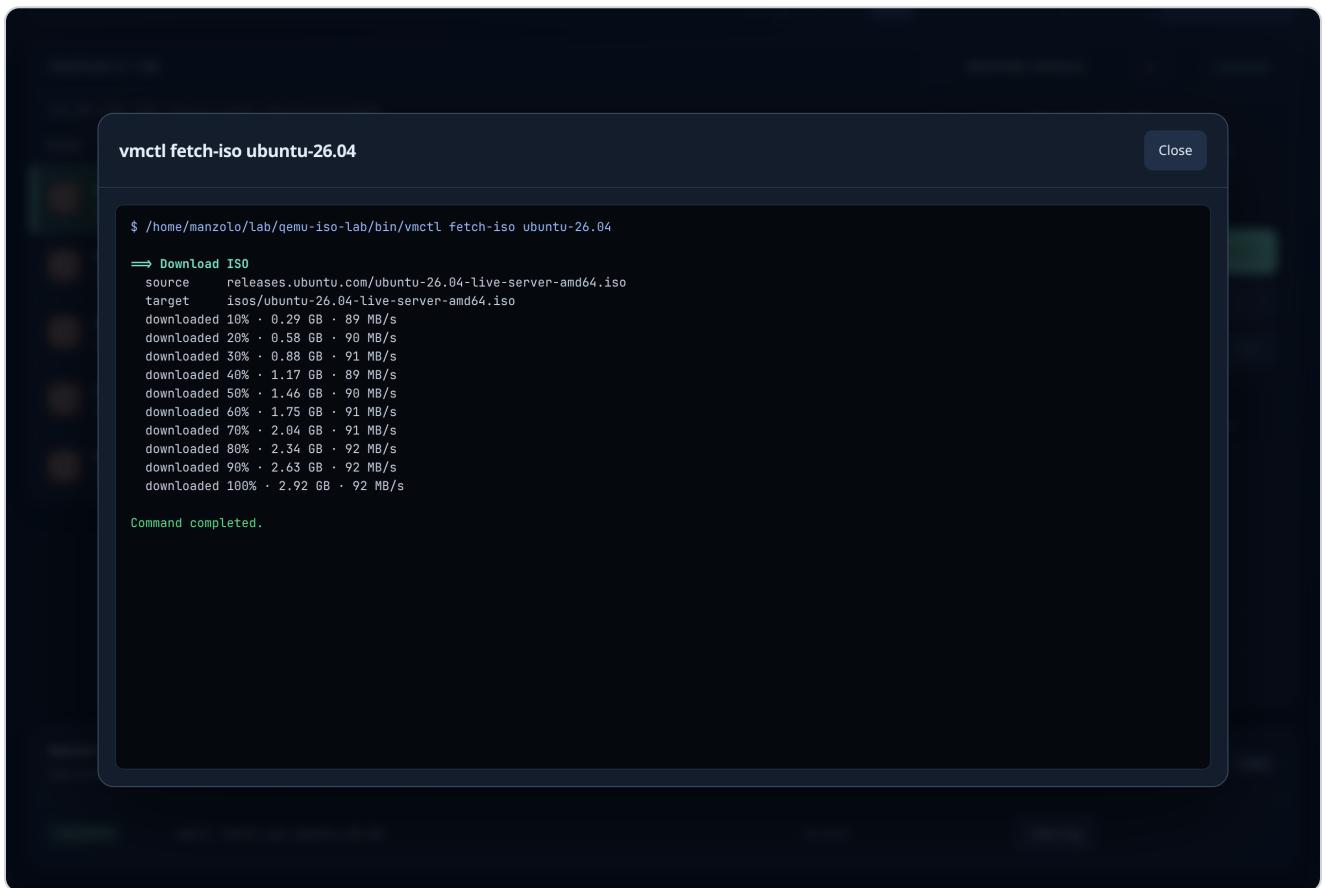
Profile	State	RAM / CPU
ubuntu-26.04 UNATTENDED Ubuntu 26.04 Desktop (unattended, autoinstall)	No disk	4G / 2
ubuntu-desktop-live MANUAL Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri UNATTENDED Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci UNATTENDED Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live MANUAL Ubuntu 26.04 Server	No disk	2G / 2

On the right, the "MACHINE DETAILS" panel for "ubuntu-26.04" is shown. It includes a "Customize" button, a "No disk" button, and several action buttons: "Unattended install" (with an "Enter" button), "Download the ISO", "Manual install (window on the host)", and "All commands for ubuntu-26.04..." (with an "F10" button). Below these are resource specifications: "4 GB Memory", "2 vCPUs", and "EFI Firmware". At the bottom, it shows "Disk: Not created yet", "Profile: 1.0.0", "ISO: Available to download", "Login: lab / lab · tracked default", and "SSH: 127.0.0.1:2275" with a "> SSH F4" button.

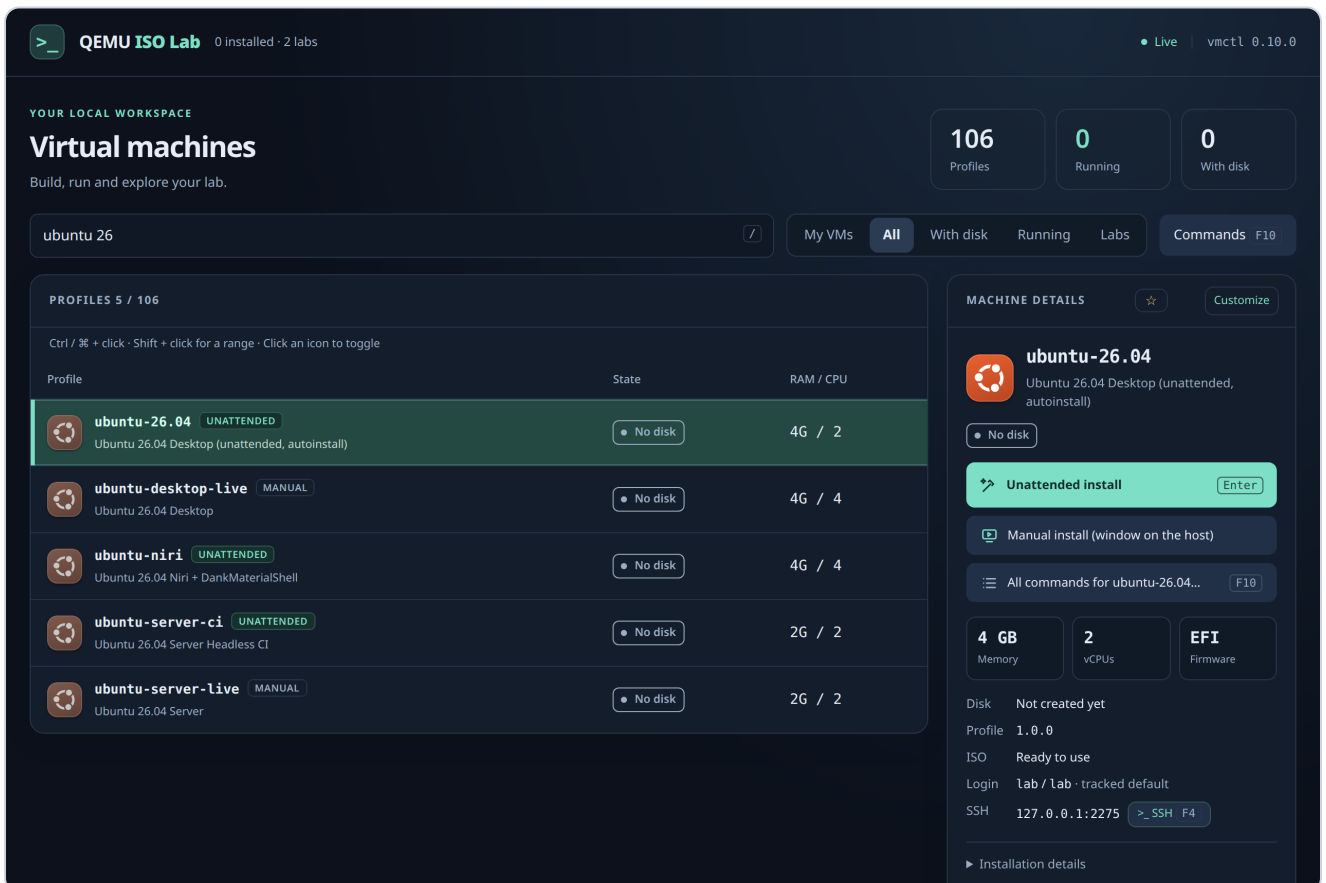
The icons tell the actions apart: green starts or installs, blue looks, amber stops, red deletes.

5. Download the ISO

Download the ISO. Every action runs as a *job*. Its log opens and follows along live, and the job keeps running if you close the log, the page or the server. The ISO is downloaded from the vendor and checked against the profile's pinned checksum.



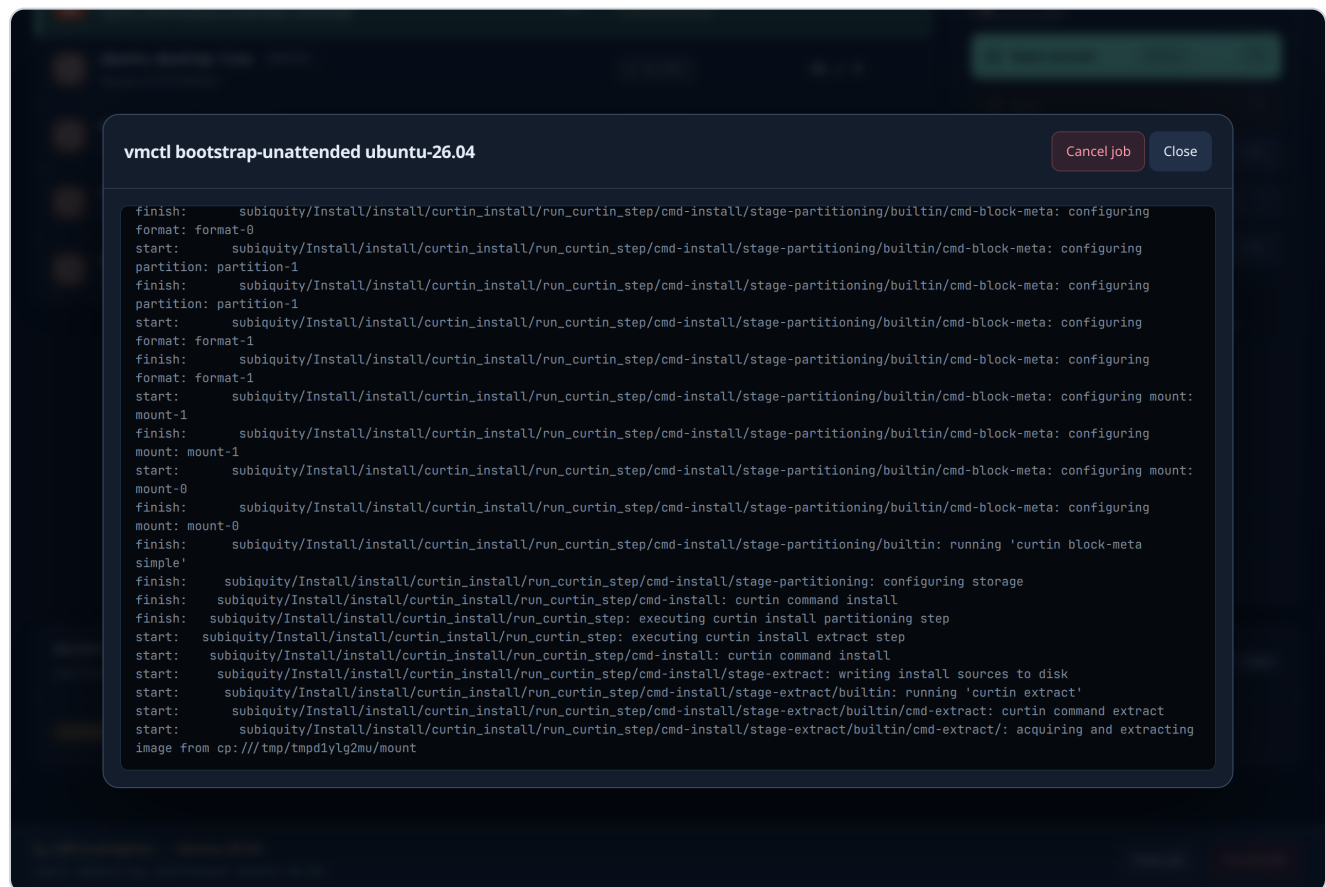
2.9 GB took about a minute here. Afterwards the panel says **ISO: Ready to use**.



Terminal equivalent: `vmctl fetch-iso ubuntu-26.04`.

6. Install it, unattended

Unattended install (or Enter). No questions and no clicks. vmctl renders the Ubuntu autoinstall answers, boots the installer headless, waits for it to finish, starts the new system and provisions it over SSH with the project's own key.



The machine shows **Installing**, and its actions become the ones that make sense during an install: the log, the console and a screenshot.

QEMU ISO Lab

1 installed · 2 labs

Live

Working...

vmctl 0.10.0

YOUR LOCAL WORKSPACE

Virtual machines

Build, run and explore your lab.

ubuntu 26

My VMs

All

With disk

Running

Labs

Commands

F10

PROFILES 5 / 106

Ctrl / ⌘ + click · Shift + click for a range · Click an icon to toggle

Profile	State	RAM / CPU
ubuntu-26.04 UNATTENDED Ubuntu 26.04 Desktop (unattended, autoinstall)	Installing	4G / 2
ubuntu-desktop-live MANUAL Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri UNATTENDED Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci UNATTENDED Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live MANUAL Ubuntu 26.04 Server	No disk	2G / 2

MACHINE DETAILS

Customize

ubuntu-26.04

Ubuntu 26.04 Desktop (unattended, autoinstall)

Installing

Open console

Enter

F3

Stop

F8

Installation log

F9

Cancel job

All commands for ubuntu-26.04...

F10

4 GB

Memory

2

vCPUs

EFI

Firmware

Disk

306M used / 35G

Job in progress... · ubuntu-26.04

vmctl bootstrap-unattended ubuntu-26.04

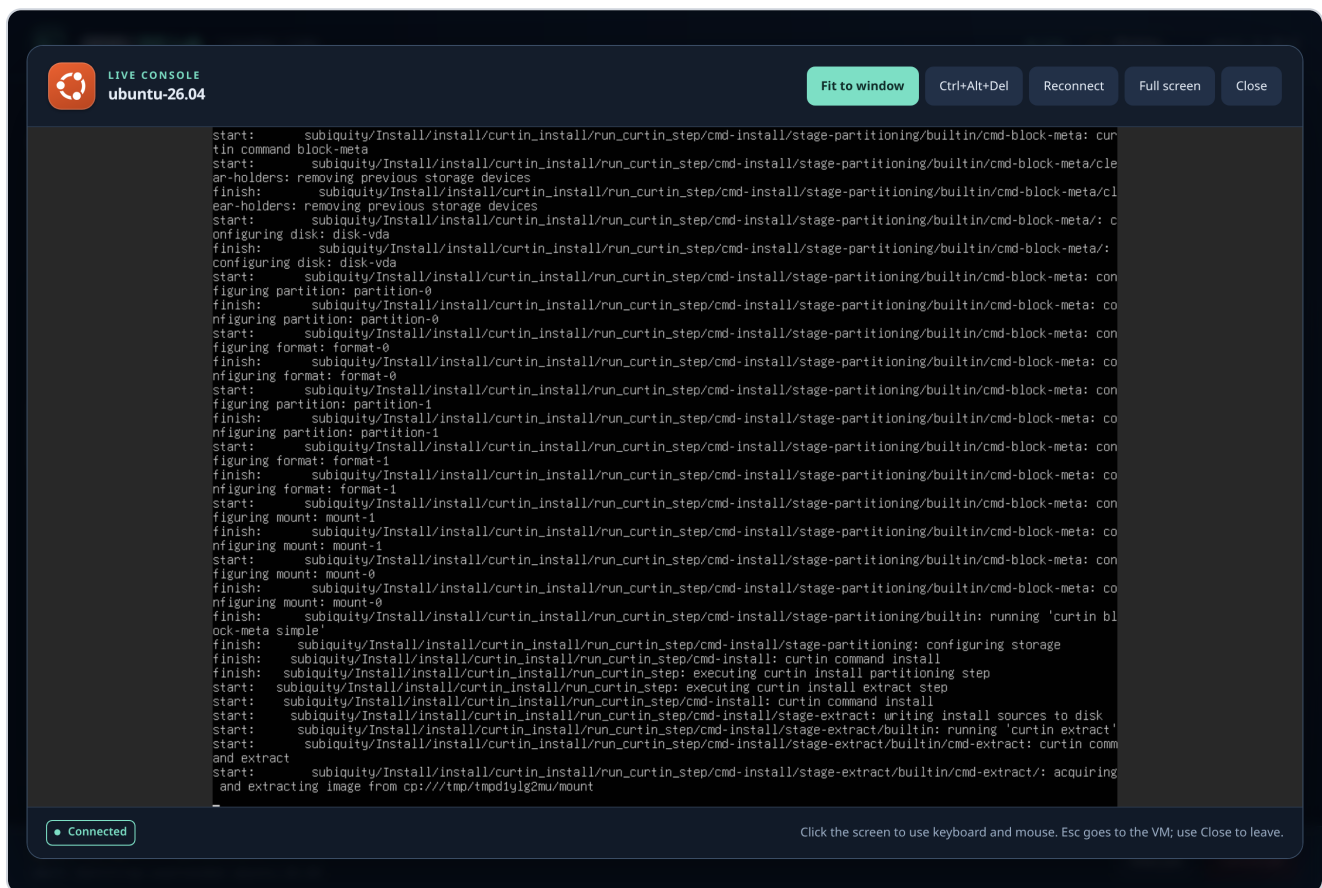
View job

Cancel job

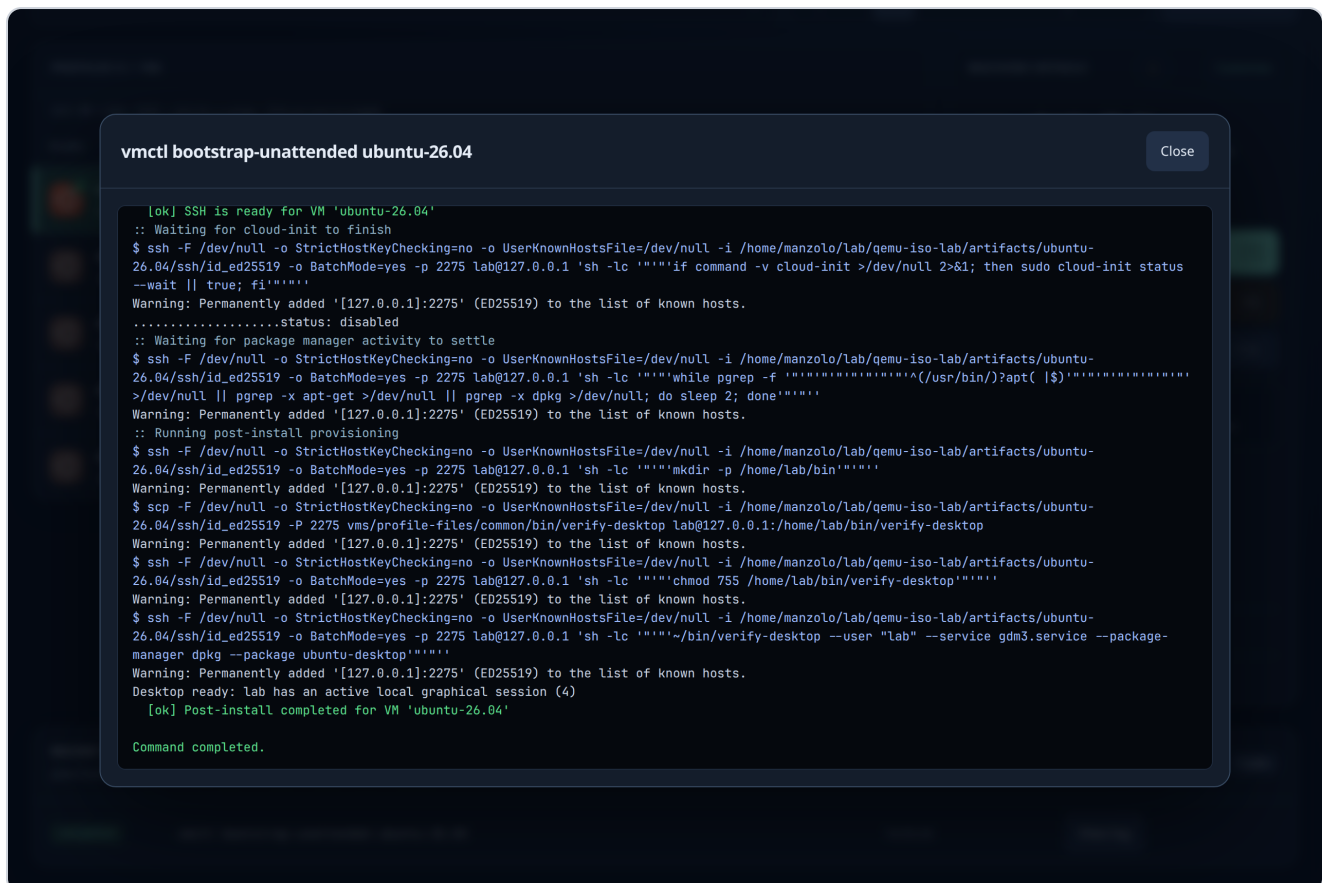
Terminal equivalent: `vmctl bootstrap-unattended ubuntu-26.04`.

7. Watch the install

Open console shows the VM's own screen in the page, with keyboard and mouse (noVNC over a WebSocket to the VM's VNC socket), even while the installer is running headless.



About 15 minutes later the log ends with the checks of the post-install: the desktop session of user lab is active, and the provisioning is complete.



The install leaves the machine **Running**, now with its everyday actions: console, screenshot, host viewer, Stop, Force stop.

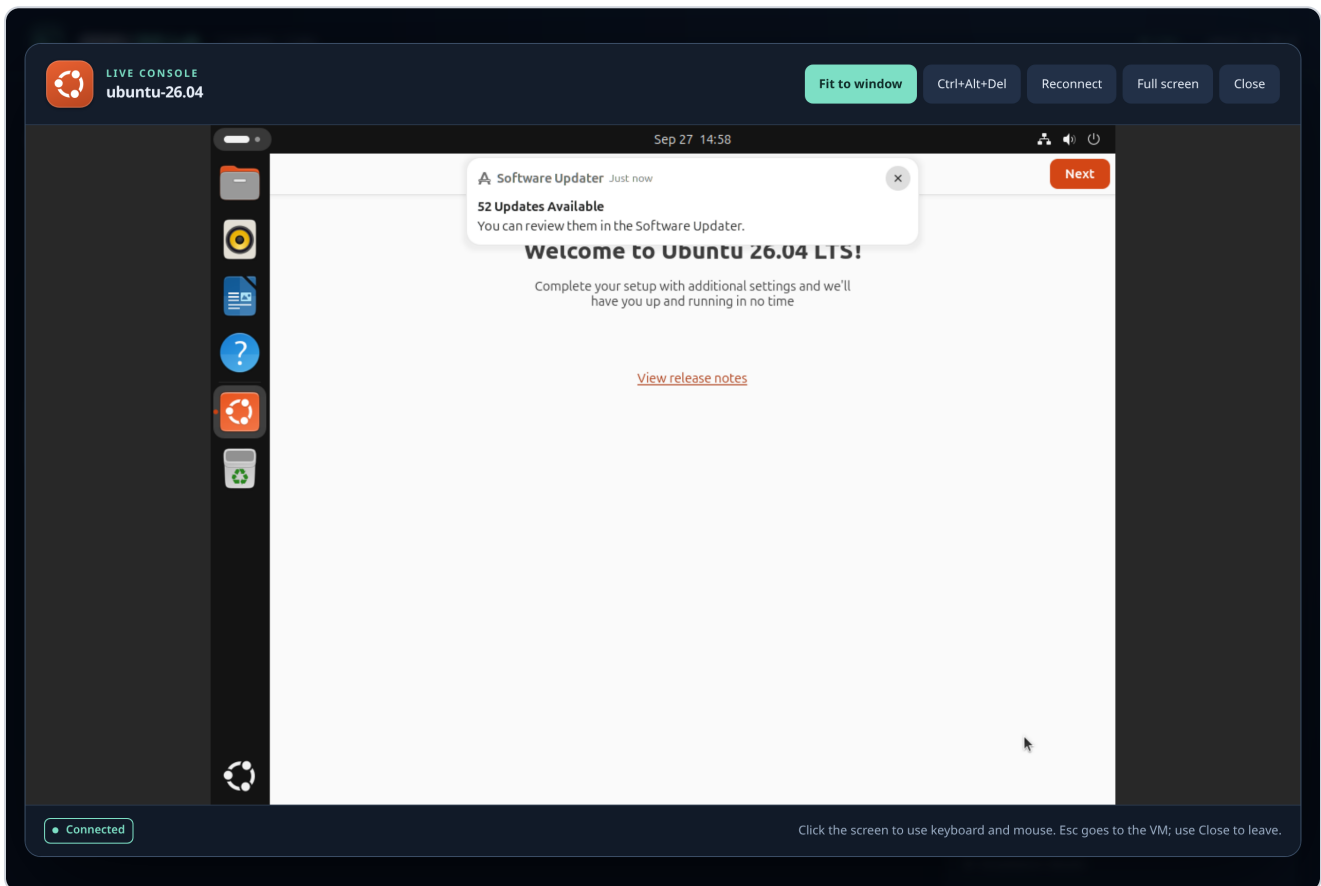
The screenshot shows the QEMU ISO Lab interface. At the top, it says "QEMU ISO Lab" with a status "1 installed · 2 labs". On the right, there are indicators for "Live", "vmctl 0.10.0", "106 Profiles", "1 Running", and "1 With disk". The main section is titled "Virtual machines" with the subtitle "Build, run and explore your lab." Below this, there's a search bar with "ubuntu 26" and filters for "My VMs", "All", "With disk", "Running", "Labs", and "Commands F10". A table lists five profiles:

Profile	State	RAM / CPU
ubuntu-26.04 (UNATTENDED) Ubuntu 26.04 Desktop (unattended, autoinstall)	Running	4G / 2
ubuntu-desktop-live (MANUAL) Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri (UNATTENDED) Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci (UNATTENDED) Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live (MANUAL) Ubuntu 26.04 Server	No disk	2G / 2

On the right, the "MACHINE DETAILS" for "ubuntu-26.04" are shown. It's an "Ubuntu 26.04 Desktop (unattended, autoinstall)" machine in a "Running" state. Controls include "Open console" (F3), "Stop" (F8), and "All commands for ubuntu-26.04..." (F10). Hardware specs are listed: 4 GB Memory, 2 vCPUs, and EFI Firmware. Other details include Disk (10.5G used / 35G), Profile (1.0.0 · on the disk), ISO (Ready to use), Login (lab / Lab · tracked default), and SSH (127.0.0.1:2275 with a terminal icon and F4 key).

8. Use the desktop

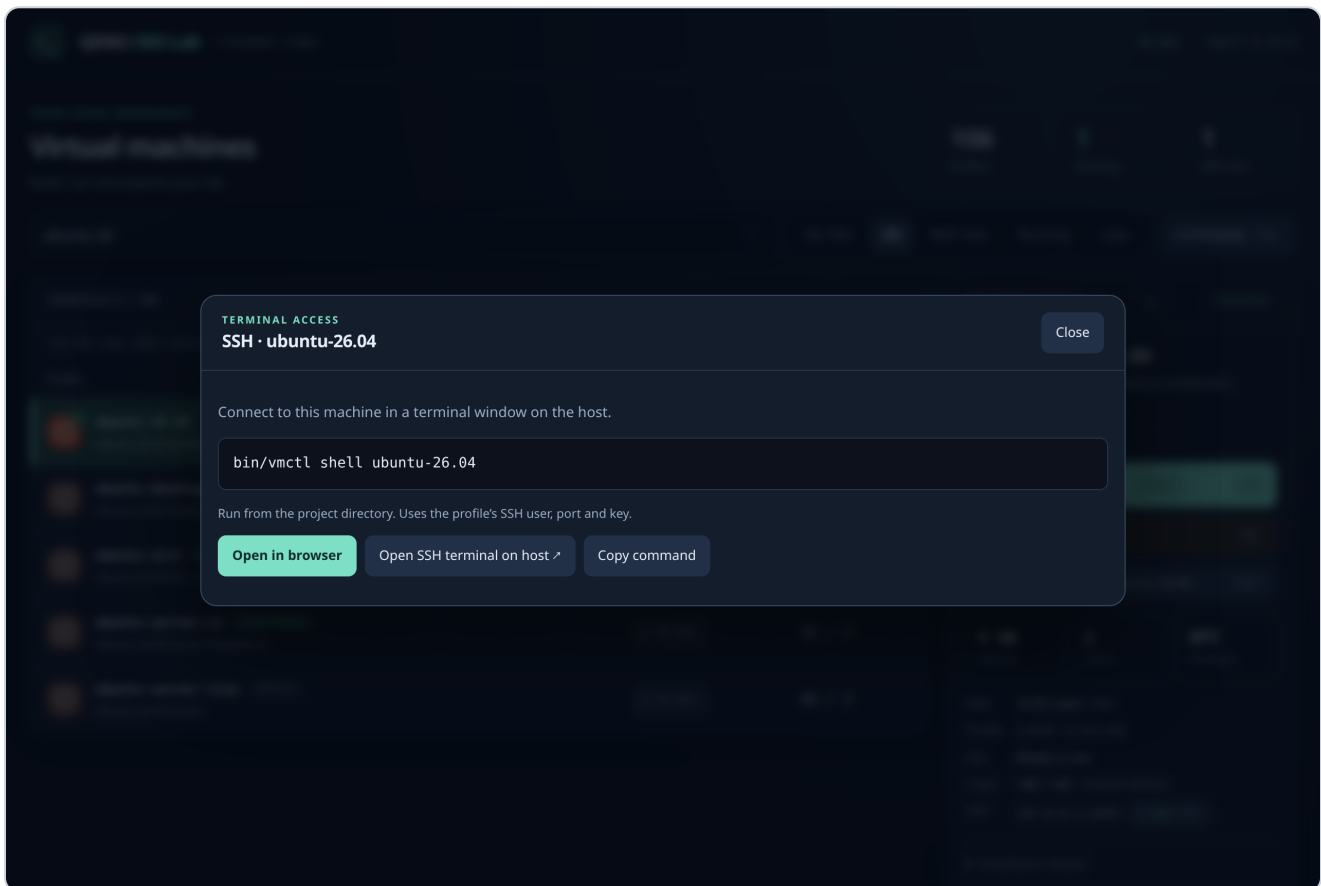
Open console again: this time it is the Ubuntu 26.04 desktop, logged in automatically. *Fit to window* / *actual size*, *Ctrl+Alt+Del* and *Full screen* are in the toolbar. *Esc* goes to the VM, and **Close** leaves the console.



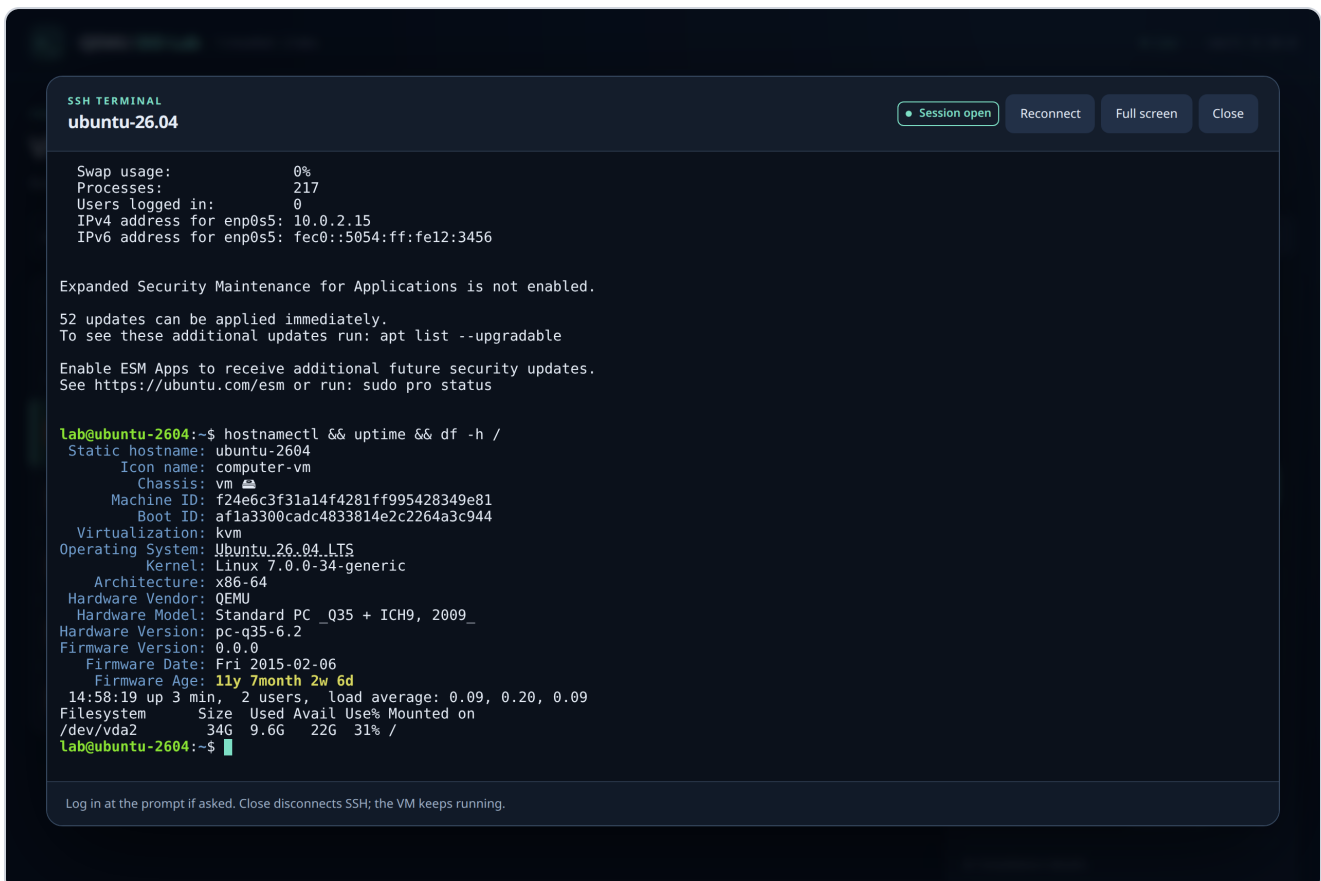
Terminal equivalent: `vmctl attach ubuntu-26.04` (a viewer window on the host).

9. SSH

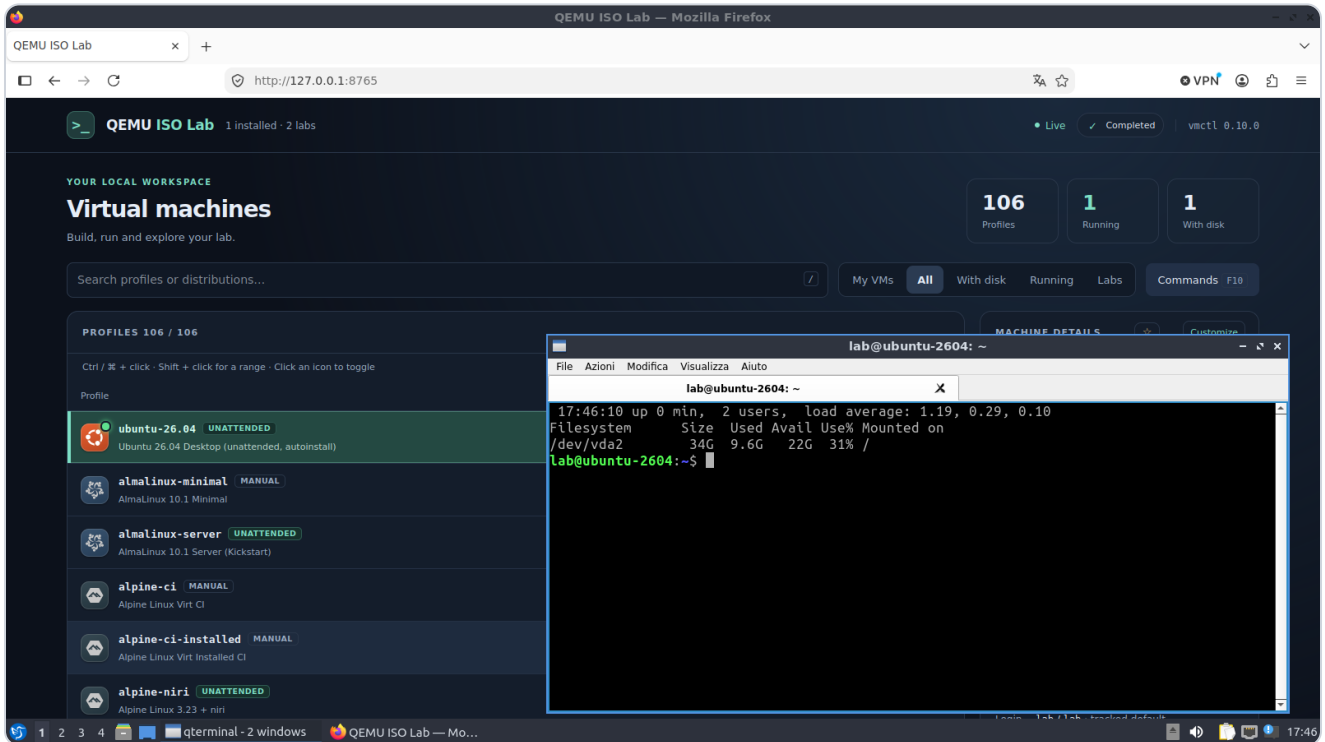
The `>_SSH` button next to the address offers three ways in, all with the profile's own key and user: in the browser, in a terminal window on the host, or the command to copy.



Open in browser is an interactive terminal in the page (xterm.js over a PTY). Closing it ends only the SSH session, never the VM.



Open SSH terminal on host opens the same session in a terminal window on the desktop.



Terminal equivalent: `vmctl shell ubuntu-26.04`.

10. Save it, stop it

Right-click a machine (or Shift+F10) for the same actions anywhere in the list, plus SSH and **Customize** (RAM, vCPUs and other settings in your own `local.json`; the catalog stays untouched).

QEMU ISO Lab 1 installed · 2 labs Live vmctl 0.10.0

Virtual machines

Build, run and explore your lab.

ubuntu 26 / My VMs **All** With disk Running Labs Commands F10

PROFILES 5 / 106

Ctrl / ⌘ + click · Shift + click for a range · Click an icon to toggle

Profile	State	RAM / CPU
ubuntu-26.04 UNATTENDED Ubuntu 26.04 Desktop (unattended, autoinstall)	Running	4G / 2
ubuntu-desktop-live MANUAL Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri UNATTENDED Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci UNATTENDED Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live MANUAL Ubuntu 26.04 Server	No disk	2G / 2

MACHINE DETAILS ☆ Customize

ubuntu-26.04
Ubuntu 26.04 Desktop (unattended, autoinstall)

Running

Open console Enter F3

Stop F8

All commands for ubuntu-26.04... F10

4 GB Memory **2** vCPUs **EFI** Firmware

Disk 10.5G used / 35G
Profile 1.0.0 · on the disk
ISO Ready to use
Login lab / Lab · tracked default
SSH 127.0.0.1:2275 > SSH F4

► Installation details

Stop asks the guest to power off (guest agent, then ACPI) and waits for it: the disk is kept.

QEMU ISO Lab 1 installed · 2 labs Live Stopping... vmctl 0.10.0

Virtual machines

Build, run and explore your lab.

ubuntu 26 / My VMs **All** With disk Running Labs Commands F10

PROFILES 5 / 106

Ctrl / ⌘ + click · Shift + click for a range · Click an icon to toggle

Profile	State	RAM / CPU
ubuntu-26.04 UNATTENDED Ubuntu 26.04 Desktop (unattended, autoinstall)	Stopping	4G / 2
ubuntu-desktop-live MANUAL Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri UNATTENDED Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci UNATTENDED Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live MANUAL Ubuntu 26.04 Server	No disk	2G / 2

MACHINE DETAILS ☆ Customize

ubuntu-26.04
Ubuntu 26.04 Desktop (unattended, autoinstall)

Stopping

Open console Enter F3

Stop F8

View operation log F9

Force stop

All commands for ubuntu-26.04... F10

Waiting for the guest to shut down. Force stop powers it off immediately.

4 GB Memory **2** vCPUs **EFI** Firmware

Stopping... · ubuntu-26.04 vmctl stop ubuntu-26.04 View job Cancel job

Stopped, the machine is back to **Boot verified**, with its boot actions and **Checkpoint now**.

>_

QEMU ISO Lab

1 installed · 2 labs

• Live

✓ Completed

vmctl 0.10.0

YOUR LOCAL WORKSPACE

Virtual machines

Build, run and explore your lab.

ubuntu 26

My VMsAllWith diskRunningLabsCommandsF10

PROFILES 5 / 106

Ctrl / ⌘ + click · Shift + click for a range · Click an icon to toggle

Profile	State	RAM / CPU
ubuntu-26.04 UNATTENDED Ubuntu 26.04 Desktop (unattended, autoinstall)	Stopped	4G / 2
ubuntu-desktop-live MANUAL Ubuntu 26.04 Desktop	No disk	4G / 4
ubuntu-niri UNATTENDED Ubuntu 26.04 Niri + DankMaterialShell	No disk	4G / 4
ubuntu-server-ci UNATTENDED Ubuntu 26.04 Server Headless CI	No disk	2G / 2
ubuntu-server-live MANUAL Ubuntu 26.04 Server	No disk	2G / 2

MACHINE DETAILS

☆Customize

ubuntu-26.04
Ubuntu 26.04 Desktop (unattended, autoinstall)

Stopped

Boot headless Enter F2

Boot with display (window on the host)

All commands for ubuntu-26.04... F10

4 GB
Memory

2
vCPUs

EFI
Firmware

Disk 10.5G used / 35G

Profile 1.0.0 · on the disk

ISO Ready to use

Login lab / Lab · tracked default

SSH 127.0.0.1:2275 >_ SSH F4

▶ Installation details

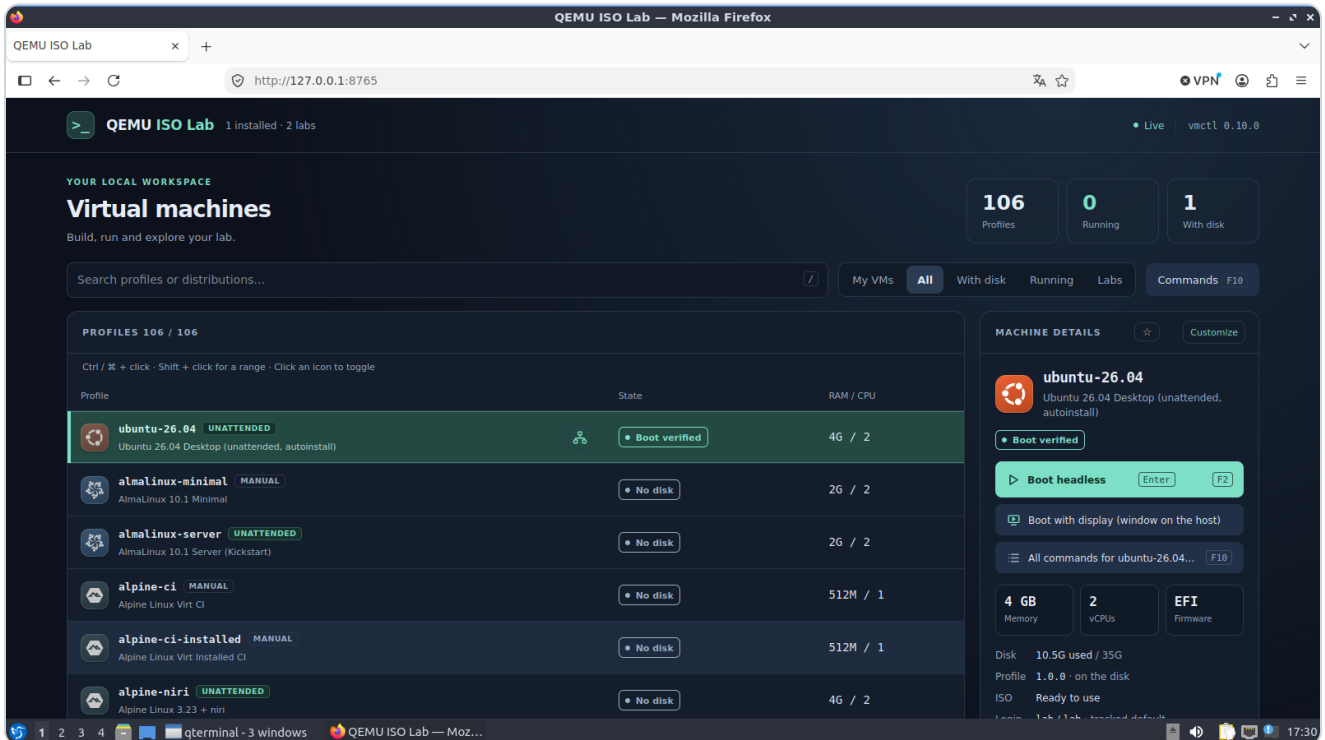
A checkpoint is a full copy of the disk (and of the EFI variables), which `vmctl clean` leaves alone. Restore it before a risky change, or keep a clean copy.

vmctl checkpoint create ubuntu-26.04 web-202609271505

Close

(75.55/100%)
(76.57/100%)
(77.58/100%)
(78.58/100%)
(79.58/100%)
(80.59/100%)
(81.59/100%)
(82.59/100%)
(83.60/100%)
(84.62/100%)
(85.63/100%)
(86.63/100%)
(87.65/100%)
(88.65/100%)
(89.67/100%)
(90.67/100%)
(91.68/100%)
(92.68/100%)
(93.69/100%)
(94.71/100%)
(95.72/100%)
(96.72/100%)
(97.74/100%)
(98.74/100%)
(99.74/100%)
(100.00/100%)
(100.00/100%)
[ok] Checkpoint 'web-202609271505' written for 'ubuntu-26.04'
Command completed.

Terminal equivalents: `vmctl stop ubuntu-26.04`, `vmctl checkpoint create ubuntu-26.04 clean`.



What is on disk

Everything lives in the repository directory, per machine:

<code>isos/ubuntu-26.04-live-server-amd64.iso</code>	the downloaded, checksummed ISO (shared by profiles)
<code>artifacts/ubuntu-26.04/</code>	
<code>— disk.qcow2, OVMF_VARS.fd</code>	the machine's disk and EFI variables
<code>— ssh/id_ed25519</code>	the key vmctl shell and the page use
<code>— logs/</code>	installer and serial console logs
<code>— runtime/tui-job/output.log</code>	the log of the last job (web page and
<code>vmtui share it)</code>	
<code>— checkpoints/<name>/</code>	the checkpoints
<code>— state.json</code>	what was installed, when, and whether its
<code>boot was verified</code>	

`vmctl clean ubuntu-26.04` deletes the disk (checkpoints stay, `--checkpoints` removes them too). `vmtui` shows the same machines, states and jobs in the terminal, and `vmctl --help` lists every command.

Next: [the web dashboard in detail](#) · [labs of several VMs](#) · [unattended flows](#) · [your own settings](#)

gemu-iso-lab · generated on 2026-10-05 from docs/FIRST_VM.md · https://github.com/manzolo/qemu-iso-lab/blob/main/docs/FIRST_VM.md